

J per goal

One goal

173.5 J → 752.9 J

4.34x avg, 26.48x peak (reasoning)

3,300+ runs, Intel x86 & GB10

A-LEMS -- MEASURED

W

One node, time resolved

4.47 W → 13.8 W

retry spike, ≤100 ms rise

runs 769/223, domain-7 trace

A-LEMS -- MEASURED

W

One rack, 32-task proxy

40.7 W → 186.8 W

~4.59x, five-seed avg.

pre-rack sim., real node traces

A-LEMS -- SIMULATED

MW per second

One facility

100 kW-1 MW/rack

ERCOT PGRR144: 10 MW/5s ceiling

vs. ceiling: EMT extension, Mo. 4-5

vs. ERCOT PGRR144 (public)

Hz

ERCOT 138 kV interface

frequency & voltage

ERCOT record: freq. to 60.235 Hz (2022)

1 Hz precedent, 2023; NERC L3 Alert

PSCAD run, Month 4-5